

| Item No. | Category       | Component                     | Description                                                        | Qty | Unit Price (₹) | Total (₹) | Supplier       | Priority  | Status   | Datasheet / Link                                                                                                                                                                                                  | Remarks                                                                                                                                                                                                                                                                                                        |
|----------|----------------|-------------------------------|--------------------------------------------------------------------|-----|----------------|-----------|----------------|-----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Controller     | ESP32 DevKit                  | Main MCU                                                           | 1   | 583            | 583       | Embetronix     | Essential | Brought  | <a href="https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32/esp-dev-kits-en-master-esp32.pdf">https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32/esp-dev-kits-en-master-esp32.pdf</a> |                                                                                                                                                                                                                                                                                                                |
| 2        | Camera Module  | ESP32-CAM                     | ESP32 Camera Development WiFi+Bluetooth Module with OV2640, OV3660 | 1   | 675            | 675       | Quarts Compone | Essential | Ordered  |                                                                                                                                                                                                                   | 2.0 Megapixel Color CMOS UXGA technology<br>8-bit compressed data output with RGB565/555 support<br>UXGA/SXGA at 15 fps<br>SVGA at 30 fps<br>CIF at 60 fps<br>1600 x 1200 resolution of UXGA<br>3590um x 2684um image area.<br>3.6V ~ 5V supply voltage                                                        |
| 3        | 2WD Kit        | Dual shaft-Straight, BO motor | DC geared motor                                                    | 2   |                | 422       | Robu           | Essential | Received |                                                                                                                                                                                                                   | Operating Voltage (V): 3-6V DC<br>Motor Rated RPM No Load Speed (3V) 90RPM +/-10%<br>Gear Reduction Ratio: 01:48<br>Motor Rated Current No Load Current (3V) Less Than 150mA                                                                                                                                   |
|          |                | Chasis                        |                                                                    |     |                |           |                |           | Received |                                                                                                                                                                                                                   | Outline Size: 135 x 135 x 80mm<br>The round acrylic plate diameter: 130mm<br>Another acrylic plate size: 135 x 75mm                                                                                                                                                                                            |
|          |                | Wheel                         |                                                                    |     |                |           |                |           | Received |                                                                                                                                                                                                                   | Plastic with Rubber tyre                                                                                                                                                                                                                                                                                       |
| 4        | Motor Driver   | TB6612FNG                     | Dual H-Bridge Driver                                               | 1   |                | 470       | Robu           | Essential | Received | <a href="https://robu-prod.com/cdn-spark">https://robu-prod.com/cdn-spark</a>                                                                                                                                     | CW/CCW/short break/stop motor control modes,<br>Built-in thermal shutdown circuit and a low voltage detecting circuit,<br>Standby control to save power,<br>CW/CCW/short brake/stop motor control modes,<br>Operating Voltage(VDC): 15, Peak Current (A): 3.2, Continuous Current (A): 1.2, No. of Channels: 1 |
| 5        | Battery        | 18650 Cell                    | LG INR18650MJ1 3400mAh (3c) LI-ION BATTERY                         | 2   | 392            | 784       | Robu           | Essential | Received | <a href="https://robu-prod.com">https://robu-prod.com</a>                                                                                                                                                         | Capacity (mAh): 3400<br>Output Voltage: 3.7V<br>Charge Rate: 3C                                                                                                                                                                                                                                                |
| 6        | Battery Holder | 2S Holder                     | Black Plastic Storage Box                                          | 1   | 17             | 17        | Robu           | Essential | Received |                                                                                                                                                                                                                   | Wire Length: 15 cm.<br>Output: 7.2v.<br>Dimension (Approx.) : 76 x 40 x 20 mm                                                                                                                                                                                                                                  |
| 7        | Charger        | USB-C 2S Charger              | 18650 Polymer Lithium-Ion Type C to 2S 8.4V 1A Booster Module      | 1   | 305            | 305       | Robu           | Essential | Received |                                                                                                                                                                                                                   | Battery Cell Composition : 2S<br>Output Voltage (V) : 8.4<br>Output Current (A) : 1<br>USB Type : TYPE - C<br>Compatibility : Li-ION 3.7V Batteries (18650, 14500)                                                                                                                                             |
| 8        | Power          | LM2596 Buck                   | LM2596S DC-DC Buck Converter Power Supply                          | 1   | 42             | 42        | Quarts Compone | Essential | Ordered  | <a href="https://robu-prod.com">https://robu-prod.com</a>                                                                                                                                                         | Input voltage: 3-40V<br>Output voltage: 1.5-35V(Adjustable)<br>Output current: Rated current is 2A, maximum 3A<br>Switching Frequency: 150KHz<br>Operating temperature: Industrial grade (-40 to +85 )<br>Conversion efficiency: 92%(highest)                                                                  |
| 9        | Switch         | Rocker Switch                 | Main Power Switch 6A 250V AC SPST ON-OFF Rocker Switch             | 1   | 16             | 16        | Quarts Compone | Essential | Ordered  |                                                                                                                                                                                                                   | Rating: Max 6A at 250VAC and 10A at 125V<br>Toggle Mode: ON-OFF<br>Structure: SPST<br>Panel Cut-Out: 18.5 x 13 mm                                                                                                                                                                                              |